

Nykaa – Beauty Store E-Commerce Platform

PRODUCT AND CATEGORY MANAGEMENT

1. Database Design

1.1 Category Table

Attribute	Data Type	Constraint
CATEGORY_ID	INT	Primary Key
CATEGORY_NAME	VARCHAR(100)	NOT NULL, UNIQUE
DESCRIPTION	VARCHAR(200)	—

1.2 Product Table

Attribute	Data Type	Constraint
PRODUCT_ID	INT	Primary Key
PRODUCT_NAME	VARCHAR(100)	NOT NULL
CATEGORY_ID	INT	Foreign Key
PRICE	NUMBER(10,2)	NOT NULL
STOCK_QUANTITY	INT	NOT NULL

2. Primary and Foreign Key Relationship

Category → Product

- Category.Category_ID → Primary Key
- Product.Product_ID → Primary Key
- Product.Category_ID → Foreign Key referencing Category.Category_ID
- Relationship: One-to-Many (1:M)

3. Table Creation

3.1 Create Category Table

```
SQL> CREATE TABLE Category (  
2     Category_ID INT PRIMARY KEY,  
3     Category_Name VARCHAR(100) NOT NULL UNIQUE,  
4     Description VARCHAR(200)  
5 );  
  
Table created.
```

3.2 Create Product Table

```
SQL> CREATE TABLE Product (  
2     Product_ID INT PRIMARY KEY,  
3     Product_Name VARCHAR(100) NOT NULL,  
4     Category_ID INT,  
5     Price NUMBER(10,2) NOT NULL,  
6     Stock_Quantity INT NOT NULL,  
7     FOREIGN KEY (Category_ID) REFERENCES Category(Category_ID)  
8 );  
  
Table created.
```

4. Data Insertion

4.1 Category Records

```
SQL> INSERT INTO Category VALUES  
2 (1, 'Skincare', 'Skin care and face care products');  
  
1 row created.  
  
SQL> INSERT INTO Category VALUES  
2 (2, 'Makeup', 'Makeup and cosmetic products');  
  
1 row created.  
  
SQL> INSERT INTO Category VALUES  
2 (3, 'Haircare', 'Hair care and hair styling products');  
  
1 row created.  
  
SQL> INSERT INTO Category VALUES  
2 (4, 'Fragrance', 'Perfumes and body fragrances');  
  
1 row created.
```

4.2 Product Records

```
SQL> INSERT INTO Product VALUES
  2 (101, 'Face Wash', 1, 299.00, 50);

1 row created.

SQL> INSERT INTO Product VALUES
  2 (102, 'Moisturizer', 1, 499.00, 30);

1 row created.

SQL> INSERT INTO Product VALUES
  2 (103, 'Lipstick', 2, 699.00, 40);

1 row created.

SQL> INSERT INTO Product VALUES
  2 (104, 'Shampoo', 3, 399.00, 25);

1 row created.

SQL> INSERT INTO Product VALUES
  2 (105, 'Perfume', 4, 899.00, 20);

1 row created.
```

5. Data Retrieval

5.1 Display Category Records

```
SQL> SELECT * FROM Category;

CATEGORY_ID
CATEGORY_NAME
-----
DESCRIPTION
-----
1
Skincare
Skin care and face care products

2
Makeup
Makeup and cosmetic products

CATEGORY_ID
CATEGORY_NAME
-----
DESCRIPTION
-----
3
Haircare
Hair care and hair styling products

4
Fragrance
```

```

CATEGORY_ID
-----
CATEGORY_NAME
-----
DESCRIPTION
-----
Perfumes and body fragrances

```

5.2 Display Product Records

```

SQL> SELECT * FROM Product;

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID      PRICE STOCK_QUANTITY
-----
101
Face Wash        1      299           50
102
Moisturizer      1      499           30
PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID      PRICE STOCK_QUANTITY
-----
103
Lipstick         2      699           40
104
Shampoo

```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID      PRICE STOCK_QUANTITY
-----
3                399           25
105
Perfume          4      899           20

```

6. Product Update

```
SQL> UPDATE Product
  2  SET Price = 349.00,
  3      Stock_Quantity = 60
  4  WHERE Product_ID = 101;

1 row updated.

SQL> UPDATE Product
  2  SET Price = 549.00
  3  WHERE Product_ID = 102;

1 row updated.
```

7. Product Delete

```
SQL> DELETE FROM Product
  2  WHERE Product_ID = 105;

1 row deleted.

SQL> DELETE FROM Product
  2  WHERE Product_ID = 103;

1 row deleted.
```

8. Category-wise Product Reports

8.1 Product Listing by Category

This report displays the category name along with the products belonging to each category.

```
SQL> SELECT
  2     C.Category_Name,
  3     P.Product_Name,
  4     P.Price,
  5     P.Stock_Quantity
  6 FROM Category C
  7 JOIN Product P
  8 ON C.Category_ID = P.Category_ID
  9 ORDER BY C.Category_Name;
```

CATEGORY_NAME

PRODUCT_NAME

PRICE STOCK_QUANTITY

Haircare

Shampoo

399

25

Skincare

Face Wash

349

60

CATEGORY_NAME

PRODUCT_NAME

PRICE STOCK_QUANTITY

PRICE STOCK_QUANTITY

Skincare

Moisturizer

549

30

8.2 Product Listing By Specific Category

```
SQL> SELECT
  2     P.Product_ID,
  3     P.Product_Name,
  4     P.Price,
  5     P.Stock_Quantity
  6 FROM Product P
  7 JOIN Category C
  8 ON P.Category_ID = C.Category_ID
  9 WHERE C.Category_Name = 'Skincare';
```

PRODUCT_ID

PRODUCT_NAME

	PRICE	STOCK_QUANTITY
101		
Face Wash	349	60
102		
Moisturizer	549	30

PRODUCT_ID

PRODUCT_NAME

	PRICE	STOCK_QUANTITY
--	-------	----------------

8.3 Product Count by Category

```
SQL> SELECT
  2     C.Category_Name,
  3     COUNT(P.Product_ID) AS Product_Count
  4 FROM Category C
  5 LEFT JOIN Product P
  6 ON C.Category_ID = P.Category_ID
  7 GROUP BY C.Category_Name;
```

CATEGORY_NAME

PRODUCT_COUNT

Skincare	2
----------	---

Haircare	1
----------	---

Makeup	0
--------	---

CATEGORY_NAME

PRODUCT_COUNT

Fragrance	0
-----------	---